

GOOGLE CYBERSECURITY CERTIFICATE

Wireshark Labs

Google Cybersecurity Professional Certificate · Study Guide

Packet analysis with Wireshark's GUI — filtering by IP and MAC address, inspecting protocol layers inside a packet, examining DNS queries/answers, and reading a TCP/HTTP log to recognize a SYN flood DoS attack in progress.

Track	Google Cybersecurity Professional Certificate
Course	Detection and Response
Tool	Wireshark (GUI packet analyzer)
Analyst	Chayan Panchal

1 Exploring and Filtering Packet Data

Investigating a user's web browsing session from a sample packet capture — identifying source/destination IPs, the protocols in use, and the data exchanged.

Column	Meaning
No.	Index of the packet in the capture
Time	Timestamp of the packet
Source / Destination	Source and destination IP address
Protocol	Protocol contained in the packet
Info	Wireshark's interpretation of the packet payload



Display filters used in this lab

```
ip.addr == 142.250.1.139    # traffic to/from a specific IP
ip.src == 142.250.1.139    # traffic FROM a specific IP only
ip.dst == 142.250.1.139    # traffic TO a specific IP only
eth.addr == 42:01:ac:15:e0:02 # traffic for a specific MAC address
udp.port == 53              # DNS query/response traffic
tcp.port == 80              # HTTP (port 80) traffic
tcp contains "curl"         # payload text search
```

NOTE Drilling into a single TCP packet's subtrees — Frame, Ethernet II, Internet Protocol Version 4, Transmission Control Protocol — walks through the OSI/TCP-IP layers from the physical frame up to the transport-layer flags and ports for that one packet.

Question	Answer
Protocol of the first 'Echo (ping) request' packet	ICMP
TCP destination port of the inspected TCP packet	80
Protocol for the packet matching the filtered MAC address	TCP
Domain queried in the DNS request	opensource.google.com

2 Reading a Wireshark TCP/HTTP Log — Spotting a SYN Flood

A separate TCP/HTTP log for traffic between employee visitors and the company's web server (192.0.2.1) shows the normal three-way handshake, then a SYN flood DoS attack developing from a single attacker IP (203.0.113.0).



Normal TCP Three-Way Handshake

1. [SYN]

Client requests a connection

2. [SYN, ACK]

Server agrees, reserves resources

3. [ACK]

Client confirms — connection established



A normal, successful visitor transaction

```
47 3.144521 198.51.100.23 -> 192.0.2.1 TCP [SYN]
48 3.195755 192.0.2.1 -> 198.51.100.23 TCP [SYN, ACK]
49 3.246989 198.51.100.23 -> 192.0.2.1 TCP [ACK]
50 3.298223 198.51.100.23 -> 192.0.2.1 HTTP GET /sales.html HTTP/1.1
51 3.349457 192.0.2.1 -> 198.51.100.23 HTTP HTTP/1.1 200 OK
```



Attacker (203.0.113.0) floods SYN packets, never completes the handshake

```
52 3.390692 203.0.113.0 -> 192.0.2.1 TCP [SYN] <- attacker
57 3.664863 203.0.113.0 -> 192.0.2.1 TCP [SYN] <- repeats, no ACK ever sent
59 3.795332 203.0.113.0 -> 192.0.2.1 TCP [SYN]
61 3.939499 203.0.113.0 -> 192.0.2.1 TCP [SYN]
...(SYN packets from 203.0.113.0 continue every ~0.3s for 25+ seconds)...
77 7.330577 192.0.2.1 -> 198.51.100.5 TCP HTTP/1.1 504 Gateway Time-out
121 19.844731 192.0.2.1 -> 198.51.100.9 TCP [RST, ACK] <- legit visitors dropped
```

IMPACT The attacker's SYN requests reserve server resources without ever finishing the handshake, exhausting capacity for legitimate connections. Real visitors start seeing 504 Gateway Time-out errors and [RST, ACK] resets, and by log entry 125 the server stops responding to anyone but the attacker — a single source IP makes this a direct (not distributed) DoS SYN flood.

3 Quick Reference



Wireshark — quick reference

```
ip.addr == X / ip.src == X / ip.dst == X      # filter by IP
eth.addr == XX:XX:XX:XX:XX:XX                # filter by MAC
udp.port == 53                                # DNS traffic
tcp.port == 80                                # HTTP traffic
tcp contains "text"                           # search payload text
```

TCP handshake: [SYN] -> [SYN, ACK] -> [ACK]

SYN flood tell: many [SYN] from one source IP, no completed [ACK], followed by
[RST, ACK] resets and 504 Gateway Time-out for legitimate traffic